FLATNESS OF A MAP OF THICKENINGS PROPAGATES FROM THE REDUCTION WHEN THE SOURCE IS FLAT FINITELY PRESENTED

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4247-4265 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 4267-4288, the complete author proof (20 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side (only B has a complete proof).

Editorial note. This lemma is the fp refinement our wave-13/16 thickening-flatness cards cite by name; reviving it closes that pointer.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. More on Morphisms, lemma flatness-morphism-thickenings-fp-over-ft (schemes engine); standard etale-locality of flatness; Properties of Spaces chart existence.

Source reference. Original lemma 0CF3: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4288–4288. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Consider a commutative diagram*

$$\begin{array}{ccc} (X \subset X') & \xrightarrow[(f,f')]{} & (Y \subset Y') \\ & \searrow \quad \swarrow & \\ & (B \subset B') & \end{array}$$

of thickenings of algebraic spaces over S. Assume

(1) *$Y' \to B'$ is locally of finite type,*
(2) *$X' \to B'$ is flat and locally of finite presentation,*
(3) *f is flat, and*
(4) *$X = B \times_{B'} X'$ and $Y = B \times_{B'} Y'$.*

Then f' is flat and for all $y' \in |Y'|$ in the image of $|f'|$ the morphism $Y' \to B'$ is flat at y'.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Choose a scheme U' and a surjective étale morphism $U' \to B'$. Choose a scheme V' and a surjective étale morphism $V' \to U' \times_{B'} Y'$. Choose a scheme W' and a surjective étale morphism $W' \to V' \times_{Y'} X'$. Let U, V, W be the base change of U', V', W' by $B \to B'$. Then flatness of f' is equivalent to flatness of $W' \to V'$ and we are given that $W \to V$ is flat. Hence we may apply the lemma in the case of schemes to the diagram

$$\begin{array}{ccc} (W \subset W') & \longrightarrow & (V \subset V') \\ & \searrow \quad \swarrow & \\ & (U \subset U') & \end{array}$$

of thickenings of schemes. See More on Morphisms, Lemma 0CF3. The statement about flatness of Y'/B' at points in the image of f' follows in the same manner. $\square$